Ratio, Root, and Alternating Series Tests

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Ratio Test**

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{(n+1)/2^{n+1}}{n/2^n} = \lim_{n\to\infty} \frac{n+1}{2n} = \frac{1}{2} < 1$. Converges by the Ratio Test.

2 **[Foundational]** • **Target: The Root Test**

Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.

$\lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{n}{2n+1}\right)^n\right|} = \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2} < 1$. Converges by the Root Test.

3 **[Foundational]** • **Target: Alternating series and the Alternating Series Test**

State the Alternating Series Test, and use it to show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.

The AST states $\sum(-1)^{n+1}b_n$ converges if $b_n > 0$, b_n is decreasing, and $\lim_{n\to\infty} b_n = 0$. Here $b_n = \frac{1}{n}$: positive, decreasing, and $\to 0$. The series converges (this is the alternating harmonic series).

4 **[Foundational]** • **Target: Alternating series and the Alternating Series Test**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$ converges using the Alternating Series Test.

$b_n = \frac{n}{n+1} \to 1 \neq 0$. Since b_n does not approach 0, the Alternating Series Test does not apply, and in fact the series diverges by the nth-term test.

5 **[Foundational]** • **Target: Alternating series error bound**

Approximate $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ using the first 4 terms, and bound the error using the Alternating Series Error Bound.

$S_4 = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} \approx 1 - 0.25 + 0.111 - 0.0625 = 0.7986$. Error bound: $|S - S_4| \leq b_5 = \frac{1}{25} = 0.04$.

6 **[Foundational]** • **Target: Absolute vs. conditional convergence**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges absolutely, conditionally, or diverges.

$\sum\left|\frac{(-1)^{n+1}}{n^2}\right| = \sum\frac{1}{n^2}$, a convergent p-series ($p = 2 > 1$). So the series converges ABSOLUTELY.

7 **[Foundational]** • **Target: More Ratio Test practice**

Explain why the Ratio Test is INCONCLUSIVE for $\sum_{n=1}^{\infty} \frac{1}{n}$, and identify which test should be used instead.

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{n}{n+1} = 1$. Since the limit equals exactly 1, the Ratio Test gives no conclusion. The p-series test (or Integral Test) should be used instead, correctly showing divergence.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Partial sums of an alternating series (visualizing oscillating convergence)**

Compute the first 5 partial sums of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ (the alternating harmonic series, which converges to $\ln 2 \approx 0.693$), and describe how they oscillate.

$S_1 = 1$, $S_2 = 1 - \frac{1}{2} = 0.5$, $S_3 = 0.5 + \frac{1}{3} \approx 0.833$, $S_4 \approx 0.833 - 0.25 = 0.583$, $S_5 \approx 0.583 + 0.2 = 0.783$. The partial sums oscillate above and below the true sum $\ln 2 \approx 0.693$, with each successive sum landing closer — a signature behavior of alternating series.

9 **[Intermediate]** • **Target: Determining the number of terms needed for a given accuracy**

For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$, find the minimum number of terms needed so the partial sum is within 0.001 of the true sum, using the Alternating Series Error Bound.

Need $b_{n+1} = \frac{1}{(n+1)^3} < 0.001 \Rightarrow (n+1)^3 > 1000 \Rightarrow n+1 > 10 \Rightarrow n \geq 10$. So 10 terms suffice.

10 **[Intermediate]** • **Target: Mixed practice: classify each series**

Classify each series as convergent (state absolutely or conditionally) or divergent:

a $\sum \frac{(-1)^n}{\sqrt{n}}$ Conditionally convergent — converges by AST ($b_n = \frac{1}{\sqrt{n}} \to 0$, decreasing), but $\sum \frac{1}{\sqrt{n}}$ diverges ($p = \frac{1}{2} \leq 1$)

b $\sum \frac{(-1)^n n^2}{n^3+1}$ Conditionally convergent — converges by AST, but $\sum \frac{n^2}{n^3+1}$ diverges by Limit Comparison to $\sum \frac{1}{n}$

c $\sum \frac{(-1)^n}{n!}$ Absolutely convergent — $\sum \frac{1}{n!}$ converges by the Ratio Test (ratio $\to 0$)

(a) Conditionally convergent — converges by AST ($b_n = \frac{1}{\sqrt{n}} \to 0$, decreasing), but $\sum \frac{1}{\sqrt{n}}$ diverges ($p = \frac{1}{2} \leq 1$); (b) Conditionally convergent — converges by AST, but $\sum \frac{n^2}{n^3+1}$ diverges by Limit Comparison to $\sum \frac{1}{n}$; (c) Absolutely convergent — $\sum \frac{1}{n!}$ converges by the Ratio Test (ratio $\to 0$)

11 **[Intermediate]** • **Target: Why the Ratio Test fails for p-series**

Show that the Ratio Test is inconclusive for EVERY p-series $\sum \frac{1}{n^p}$, regardless of p.

$\lim_{n\to\infty} \left|\frac{1/(n+1)^p}{1/n^p}\right| = \lim_{n\to\infty} (\frac{n}{n+1})^p = 1^p = 1$ for ANY value of p. Since the limit always equals exactly 1, the Ratio Test can never determine convergence or divergence for p-series — a different test (the p-series test itself) is required.

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12 **[Intermediate]** • **Target: Combining multiple tests in one problem**

Level 3 · Advanced / AP FRQLet $f(x)$ be a function whose third derivative satisfies $|f'''(x)| \le 6$ for all $x \in [0, 0.4]$. If the second-degree Taylor polynomial $P_2(x)$ centered at $c = 0$ is used to approximate $f(0.2)$, find the maximum possible error in this approximation using the Lagrange Error Bound.

Using the Lagrange Error Bound formula with $n = 2$, $c = 0$, $x = 0.2$, and $M = 6$: $|R_2(0.2)| \le \frac{M}{(2+1)!}|x-c|^{2+1} = \frac{6}{3!}|0.2-0|^3$. Since $3! = 6$: $|R_2(0.2)| \le \frac{6}{6}(0.2)^3 = 1 \times 0.008 = \mathbf{0.008}$ $\left(\text{or } \frac{1}{125}\right)$. The maximum truncation error is guaranteed to be no greater than 0.008.

13 **[Intermediate]** • **Target: The Ratio Test**

[Beyond AB — Core BC Topics Overview · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

14 **[Intermediate]** • **Target: The Root Test**

[Unit 9 · Parametric Equations & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Alternating series and the Alternating Series Test** • **Cross-Lesson: 2.01 Euler's Method and Logistic Growth**

Level 2 · MediumFind the minimum number of subintervals n required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than 10^{-4}.

Compute second derivative: $f(x) = x^{-1} \Rightarrow f'(x) = -x^{-2} \Rightarrow f''(x) = 2x^{-3} = \frac{2}{x^3}$. On $[1, 3]$, $|f''(x)| = \frac{2}{x^3}$ is decreasing, so its maximum occurs at $x = 1$: $K = \frac{2}{1^3} = 2$. Using the Trapezoidal error bound with $a = 1, b = 3$: $|E_T| \le \frac{K(b-a)^3}{12n^2} = \frac{2(3-1)^3}{12n^2} = \frac{2(8)}{12n^2} = \frac{16}{12n^2} = \frac{4}{3n^2}$. Require.

16 **[Challenging]** • **Target: Alternating series error bound** • **Cross-Lesson: 3.01 Parametric Equations**

Level 3 · Advanced / AP FRQEvaluate the improper integral $\int_0^\infty xe^{-2x}\,dx$ or establish that it diverges.

Rewrite with limit notation: $\lim_{b\to\infty} \int_0^b xe^{-2x}\,dx$. Use integration by parts: $u = x \Rightarrow du = dx$, $dv = e^{-2x}\,dx \Rightarrow v = -\frac{1}{2}e^{-2x}$. $\int xe^{-2x}\,dx = -\frac{1}{2}xe^{-2x} - \int\left(-\frac{1}{2}e^{-2x}\right)dx = -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}$. Evaluate from 0 to b: $\lim_{b\to\infty}\left[-\frac{1}{2}be^{-2b} - \frac{1}{4}e^{-2b} - \left(0 - \frac{1}{4}e^0\right)\right]$. By L'Hôpital's Rule.

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17 **[Challenging]** • **Target: Absolute vs. conditional convergence** • **Cross-Lesson: 3.02 Vector-Valued Functions**

Does $\sum_{n=0}^{\infty} (0.9)^n$ converge, and to what?

Method · Geometric Series Convergence & Sum: Step 1 · Identify series parameters: Initial term: $a = (0.9)^0 = 1$. Common ratio: $r = 0.9 = \frac{9}{10}$. Step 2 · Test convergence: $|r| = 0.9 < 1$, therefore the series converges. Step 3 · Calculate exact sum: $S = \frac{a}{1-r} = \frac{1}{1-0.9} = \frac{1}{0.1} = \mathbf{10}$. Conclusion: Converges to $\mathbf{10}$.

18 **[Challenging]** • **Target: More Ratio Test practice** • **Cross-Lesson: 4.01 Polar Curves and Derivatives**

Level 1 · FoundationsEvaluate the indefinite integral $\int \arctan(x)\,dx$.

Use LIATE: $u = \arctan(x) \Rightarrow du = \frac{1}{1+x^2}\,dx$, and $dv = dx \Rightarrow v = x$. Applying parts: $\int \arctan(x)\,dx = x\arctan(x) - \int \frac{x}{1+x^2}\,dx$. For the second integral, let $w = 1 + x^2 \Rightarrow dw = 2x\,dx$: $\int \frac{x}{1+x^2}\,dx = \frac{1}{2}\ln(1+x^2)$. Thus: $\mathbf{x\arctan(x) - \frac{1}{2}\ln(1+x^2) + C}$.

19 **[Challenging]** • **Target: Partial sums of an alternating series (visualizing oscillating convergence)** • **Cross-Lesson: 4.02 Area in Polar Coordinates**

Level 1 · FoundationsFor the parametric curve $x(t) = \cos(t)$ and $y(t) = \sin(2t)$, find all values of $t \in [0, \pi]$ where the curve has a vertical tangent line.

A vertical tangent occurs when $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} eq 0$. Compute $x'(t) = -\sin(t) = 0 \Rightarrow t = 0, \pi$. Check $y'(t) = 2\cos(2t)$: • At $t = 0$: $y'(0) = 2\cos(0) = 2 eq 0$. • At $t = \pi$: $y'(\pi) = 2\cos(2\pi) = 2 eq 0$. Therefore, vertical tangents occur at $\mathbf{t = 0}$ and $\mathbf{t = \pi}$.

20 **[Challenging]** • **Target: Determining the number of terms needed for a given accuracy** • **Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests**

Level 2 · MediumA particle moves with position vector $\vec{r}(t) = \langle t^2 - 4t, \frac{1}{3}t^3 - 4t \rangle$. At what time $t > 0$ is the particle at rest?

A particle is at rest if and only if its velocity vector is $\vec{0}$ (both components are simultaneously 0). • $x'(t) = 2t - 4 = 0 \Rightarrow t = 2$. • $y'(t) = t^2 - 4 = 0 \Rightarrow t = \pm 2$. Since $t > 0$, both components are zero simultaneously at $\mathbf{t = 2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Ratio Test

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

a. 3

b. $\frac{1}{3}$ ✓

c. $\frac{4}{3}$

d. $\frac{1}{4}$

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)/3^{n+1}}{n/3^n} = \frac{n+1}{3n} \to \frac{1}{3}$. Since $\frac{1}{3} < 1$, the series converges by the Ratio Test.

22 [Foundational] • Target: The Root Test

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{4n+4}{11n-7}\right)^n\right|}$.

a. $\frac{4}{11}$ ✓

b. $\frac{11}{4}$

c. $\frac{19}{22}$

d. $\frac{15}{11}$

$\sqrt[n]{\left(\frac{4n+4}{11n-7}\right)^n} = \frac{4n+4}{11n-7} \to \frac{4}{11} = \frac{4}{11}$. Since $\frac{4}{11} < 1$, converges by the Root Test.

23 [Foundational] • Target: Alternating series and the Alternating Series Test

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{10000}$?

a. 8 ✓

b. 9

c. 7

d. 10000

The error after summing up to $n = 7$ is at most the next term, $\frac{1}{40320} < \frac{1}{10000}$, while stopping one term earlier leaves a bound $\frac{1}{5040} \geq \frac{1}{10000}$. So 8 terms are needed.

24 [Foundational] • Target: Alternating series error bound

When approximating $\sin(0.3)$ by its Taylor polynomial of degree 5 about $x = 0$, what is the alternating series error bound?

a. $\frac{(0.3)^7}{(7)!}$ ✓

b. $\frac{1}{(7)!}$

c. $\frac{(0.3)^5}{(5)!}$

d. $\frac{(0.3)^7}{(3)!}$

The error bound is the absolute value of the next omitted term: $\frac{(0.3)^7}{(7)!}$.

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25 [Foundational] • Target: Absolute vs. conditional convergence

The ratio test gives $L = 1$: the test is:

a. **inconclusive ✓**

b. divergent

c. convergent

d. absolutely convergent

Applying Convergence Tests and Power Series principles, Ratio test inconclusive at $L = 1$. Therefore, the verified correct option is inconclusive.

26 [Foundational] • Target: More Ratio Test practice

Error bound: $|\cos x - 1 + x^2/2| \le x^4/24$ (from T_4 bound). This means the error at $x = 0.1$ is at most:

a. $0.0001/24$

b. $0.1^4/24$ ✓

c. $0.1^2/2$

d. $1/24$

Applying Convergence Tests and Power Series principles, $|x^4/24| = 0.0001/24$. Therefore, the verified correct option is $0.1^4/24$.

27 [Foundational] • Target: Partial sums of an alternating series (visualizing oscillating convergence)

Series $\sum_{n=1}^{\infty} (-1)^{n-1}/\sqrt{n}$ converges by alternating series test. Is it absolutely convergent?

a. Yes, $p = 1/2 < 1$ diverges

b. **No, $\sum 1/\sqrt{n}$ diverges ($p = 1/2 \le 1$)** ✓

c. Yes, $\sum 1/\sqrt{n}$ converges

d. Cannot determine

$p = 1/2 \le 1 \Rightarrow \sum 1/\sqrt{n}$ diverges: only conditionally convergent.

28 [Foundational] • Target: Determining the number of terms needed for a given accuracy

Which test determines $\sum n^2/e^n$? Answer: by ratio test $L =$

a. e

b. 1

c. $1/e < 1$ ✓

d. e^2

$L = \lim (n+1)^2/e^{n+1} \cdot e^n/n^2 = 1/e < 1 \Rightarrow$ converges.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Mixed practice: classify each series**

Series $\sum_{n=1}^{\infty} n \cdot (1/2)^n$: ratio test gives $L =$

a. 1

b. 2

c. 1/2 ✓

d. $n/2$

$|a_{n+1}/a_n| = (n+1)/(2n) \to 1/2 < 1 \Rightarrow$ converges.

30 **[Intermediate]** • **Target: The Root Test with more complex expressions**

Alternating test: $a_n = 1/n!$: does $a_n \to 0$? Does $a_{n+1} < a_n$?

a. Neither

b. **Both** $a_n \to 0$ **and** $a_{n+1} < a_n$ ✓

c. $a_n \to 0$ only

d. $a_{n+1} > a_n$

$1/n! \to 0$ and is decreasing: both conditions met.

31 **[Intermediate]** • **Target: Why the Ratio Test fails for p-series**

The integral $\int_0^1 e^{-x^2} dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

b. $1 - 1/3 \approx 0.667$ ✓

c. $1 + 1/3 \approx 1.333$

d. 0

$\int_0^1 e^{-x^2} dx \approx 1 - 1/3 + \cdots \approx 0.667.$

32 **[Intermediate]** • **Target: Combining multiple tests in one problem**

Maclaurin series for $\tan^{-1} x$ at $x = 1$ gives $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \cdots$. This converges very slowly. The 10th partial sum approximates $\pi/4$ with error less than:

a. 1/11

b. 1/21 ✓

c. 1/19

d. 0.01

Alternating series: error $\leq a_{11} = 1/21$.

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33 **[Intermediate]** • **Target: The Ratio Test**

Absolute convergence implies conditional convergence. The converse:

a. is also true

b. **is false (counterexample: alternating harmonic series) ✓**

c. requires $a_n \geq 0$

d. is true for power series

Conditional does not imply absolute.

34 **[Intermediate]** • **Target: The Root Test**

The Maclaurin series for a function $f(x)$ is given by $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$. If the first three terms of the Maclaurin series for $f(x)$ are $5 - 2x + \frac{3}{2}x^2$, what are the values of $f(0)$, $f'(0)$, and $f''(0)$?

a. $f(0) = 5, f'(0) = -2, f''(0) = 3$ ✓

b. $f(0) = 5, f'(0) = -2, f''(0) = 6$

c. $f(0) = 5, f'(0) = -1, f''(0) = 3$

d. $f(0) = 3, f'(0) = -2, f''(0) = 5$

The Maclaurin series for a function $f(x)$ is given by $f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$. We are given the first three terms of the Maclaurin series as $5 - 2x + \frac{3}{2}x^2$. Comparing the coefficients term by term: For the constant term (x^0): $f(0) = 5$. For the x term (x^1): $\frac{f'(0)}{1!} = -2 \Rightarrow f'(0) = -2$. For the x^2 term (x^2): $\frac{f''(0)}{2!} = \frac{3}{2} \Rightarrow \frac{f''(0)}{2} = \frac{3}{2} \Rightarrow f''(0) = 3$. Therefore.

35 **[Intermediate]** • **Target: Alternating series and the Alternating Series Test**

Consider the Taylor series expansion of the function $f(x) = \frac{1}{1-x}$ centered at $x = 0$. If the sum of the first three non-constant terms of the Taylor series is denoted by $S_3(x)$, and we are given that $S_3(x) = ax + bx^2 + cx^3$, find the value of $b + c$.

a. 1

b. 2

c. **3 ✓**

d. 4

The Taylor series expansion of $f(x) = \frac{1}{1-x}$ centered at $x = 0$ is given by $f(x) = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \ldots$, for $|x| < 1$. The first three non-constant terms of this series are x, x^2, and x^3. Thus, $S_3(x) = x + x^2 + x^3$. We are given that $S_3(x) = ax + bx^2 + cx^3$. By comparing the coefficients, we have $a = 1$, $b = 1$, and $c = 1$. Therefore, $b + c = 1 + 1 = 2$.

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36 [Intermediate] • Target: Alternating series error bound

The error function $\text{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$. Using $e^{-t^2} = \sum(-1)^n t^{2n}/n!$:

a. $\text{erf}(x) = \sum x^n/n!$

b. $\text{erf}(x) = \frac{2}{\sqrt{\pi}}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{n!\,(2n+1)}$ ✓

c. $\text{erf}(x) = e^{-x^2}$

d. $\text{erf}(x) = 2x/\sqrt{\pi}$

Applying Convergence Tests and Power Series principles, Integrate term-by-term. Therefore, the verified correct option is $\text{erf}(x) = \frac{2}{\sqrt{\pi}}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{n!\,(2n+1)}$.

37 [Intermediate] • Target: Absolute vs. conditional convergence

Generalised binomial: $\sqrt{1-x} = \sum_{n=0}^{\infty}\binom{1/2}{n}(-1)^n x^n$. First three terms:

a. $1 - x - x^2$

b. $1 + x/2 - x^2/8$

c. $1 - x/2 - x^2/8$ ✓

d. $1 - x/2$

Applying Convergence Tests and Power Series principles, $1 - x/2 - x^2/8 - \cdots$ Therefore, the verified correct option is $1 - x/2 - x^2/8$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More Ratio Test practice • Cross-Lesson: 4.01 Polar Curves and Derivatives

Use alternating series estimation: $|\cos 1 - 1 + 1/2| \le |-1/4!\,| =$

a. $1/12$

b. $1/120$

c. $1/6$

d. $1/24$ ✓

Applying Convergence Tests and Power Series principles, Error $\le a_4 = 1/4! = 1/24$. Therefore, the verified correct option is $1/24$.

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39 [Challenging] • Target: Partial sums of an alternating series (visualizing oscillating convergence) • Cross-Lesson: 4.02 Area in Polar Coordinates

Approximate $\int_0^1 \frac{\sin x}{x} dx$ using power series (first three terms):

a. $1 - 1/3 + 1/10$

b. $1 - 1/18 + 1/600$ ✓

c. $1 - 1/6 + 1/20$

d. $1 - 1/18$

$\sin x/x = \sum(-1)^n x^{2n}/(2n+1)!$; $\int_0^1 = \sum(-1)^n/(2n+1)(2n+1)! = 1 - 1/18 + 1/600 - \cdots$.

40 [Challenging] • Target: Determining the number of terms needed for a given accuracy • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Convergence of $\sum n^2/3^n$: by ratio test, $L =$

a. 1

b. 1/3 ✓

c. 3

d. 0

$|(n+1)^2/3^{n+1} \cdot 3^n/n^2| \to 1/3 < 1 \Rightarrow$ converges.

41 [Challenging] • Target: Mixed practice: classify each series • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Integral test convergence of $\sum_{n=1}^{\infty} 1/n(\ln n)^2$ (from $n = 2$) gives sum approximately:

a. **converges** ✓

b. diverges

c. equals e

d. equals π^2

$\int_2^{\infty} dx/(x(\ln x)^2) = 1/\ln 2 < \infty \Rightarrow$ converges.

42 [Challenging] • Target: The Root Test with more complex expressions • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Error bound for alternating series: $|S - S_n| \leq$

a. a_n

b. S_n

c. na_n

d. a_{n+1} ✓

Alternating series error bound: first omitted term a_{n+1}.

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43 [Challenging] • Target: Why the Ratio Test fails for p-series • Cross-Lesson: 1.01 Integration by Parts

Dirichlet test: $\sum a_n b_n$ converges if a_n monotone $\to 0$ and partial sums of $\sum b_n$ are:

a. convergent

b. zero

c. increasing

d. **bounded ✓**

Dirichlet: bounded partial sums + monotone $a_n \to 0$.

44 [Challenging] • Target: Combining multiple tests in one problem • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The error in using $T_n(x)$ decreases as n increases (assuming convergence). This is because the remainder $R_n \to$

a. 1

b. $f(x)$

c. $n!$

d. 0 ✓

Remainder $R_n \to 0$ for convergent Taylor series.

45 [Challenging] • Target: The Ratio Test • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Use the integral test to show $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges. The integral $\int_2^{\infty} \frac{dx}{x(\ln x)^2} =$

a. ∞

b. π

c. $\ln 2$

d. $1/\ln 2$ ✓

$u = \ln x$: $\int_{\ln 2}^{\infty} u^{-2} du = 1/\ln 2 < \infty$.

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